INCH-POUND

MIL-P-63074A (AR)
AMENDMENT 4
15 JUNE 1990
SUPERSEDING
AMENDMENT 3
15 APRIL 1988

MILITARY SPECIFICATION

PRIMER, ELECTRIC, M120
LOADING ASSEMBLING AND PACKING

This amendment forms a part of Military Specification MIL-P-63074A (AR) dated 17 December 1979, and is approved for use within the U.S. Army Armament, Munitions and Chemical Command and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 2

* 3.3.1: Delete in its entirety and substitute:

"Ignition element charge composition. The moisture content of the ignition element charge composition after drying shall not exceed 0.02 percent."

PAGE 12

* 4.4.2.6: Add to Major 102: "See Note if lacquer is used.".

Add Note:

"Torque test shall be performed while lacquer is still wet."

PAGE 13

* 4.4.2.7: Note 1: Delete in its entirety and substitute:

"Three samples minimum shall be selected from each drying oven after completion of drying cycle. If the moisture content of any sample exceeds the requirement, that quantity of material represented by the sample shall either be redried until acceptable results are achieved or rejected. The selected samples must not be dried more than any other unit which could be selected."

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PAGE 14

4.4.2.8: Critical Defect: Add:

"2 Plug missing 100% 3.2 Visual"

PAGE 19

* Add new page 19(a), paragraph 4.4.2.14. (See next page.)

PAGE 21

* 4.6.2.1: Delete in its entirety and substitute:

"Igniter charge composition. The moisture content of the igniter charge composition shall be determined as specified in Method 102.2.1 of MIL-STD-1234 except that the electric ignition element loading assembly and only a two hour dry time shall be used. The upper insulator from the electric ignition element loading assembly may be removed prior to analysis."

* 4.6.2.1.1: Delete in its entirety.

PAGE 23

4.6.10, 4.6.10.1, 4.6.10.2, 4.6.10.3, 4.6.10.4: Delete in their entirety and substitute the following:

"4.6.10 Determination of ignition charge composition. The chemical composition of the ignition charge shall be considered satisfactory if it is in accordance with the following percentages:

Component	Percentage
Potassium Chlorate and Gum Solids	46.1 ± 2.1%
Lead Thiocyanate	35.5 ± 2.0%
Charcoal	20.0 ± 1.5%

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.14	Head loading assembly	1 OF 1	9327345
			NEXT HIGHER ASSEMBLY 9296617

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Non-functioning test	4.5.4		3.5.2	4.6.5
Major					
101	Depth to electrode		Ø.40%	3.2	Gage
102	Electrical resistance test		100%	3.4	4.6.3
103	Flash hole obstructed		Ø.40%	3.2	Gage
104	Closing plug fails to withstand application of minimum assembly torque		Ø.40%	3.8.1	4.6.4.1
105	Functioning test	4.5.3		3.5.1	4.6.5
Minor					
201	Lacquer missing from closing plug threads		Ø.65%	3.2	Visual
202	Evidence of poor workmanship		Ø.65%	3.10	Visual

NOTES:

3 (19a)

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4.6.10.1 Charcoal. Transfer approximately 2 grams of the igniter charge composition into a tared medium porosity crucible. Place in oven 70°C overnight to drive off water. Place crucible in desiccator to cool and then weigh to establish sample weight. Wash with ten 25 ml portions of a 5 percent potassium nitrate solution followed by 50 ml of distilled water. Use a small glass rod to bread up any lumps in the sample during the extraction process. Preserve the filtrate for the determination for lead thiocyanate. Place in oven maintained at 100 ± 5°C for 2 hours, cool in a desiccator and weigh. Calculate the percentage of charcoal as follows:

$$\text{Percent charcoal} = \frac{(A-B)\ (100)}{A}$$

A = Beginning sample weight
B = Empty crucible weight

4.6.10.2 Lead thiocyanate. Exactly 35 ml of standardized 0.1 normal (N) silver nitrate shall be added to the filtrate reserved from the charcoal determination. The flask and contents shall be swirled and 100 ml of 3N nitric acid (Nitric acid used should be water white and free of oxides of nitrogen) shall be added. The swirling of the mixture shall be continued for five minutes, 5 ml of a saturated solution of ferric ammonium sulfate be added and the solution titrated with standardized 0.1N potassium thiocyanate solution to the first permanent pale rose coloration. The percentage of lead thiocyanate in the sample shall be calculated on a dry basis as follows:

$$\text{Percent lead thiocyanate} = \frac{16.17\ (AB-CD)}{E}$$

A = Volume of silver nitrate solution used
B = Normality of silver nitrate solution
C = Volume of potassium thiocyanate solution
D = Normality of potassium thiocyanate
E = Weight of sample

4.6.10.3 Potassium chlorate & gum solids. The percentage of potassium chlorate and gum solids is obtained by subtracting the total percentage of charcoal and lead thiocyanate from one hundred percent."

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PAGE 25

4.6.11: Delete in its entirety and substitute:

"4.6.11 Proving ground test. The primers shall be assembled into complete rounds for which the primer is standard and fired from an M60 or M60A1 tank (or from any appropriate facility mount) with an M68 gun in accordance with ATP IS-P-7J. The test shall be fired with a standard service charge at 145°F -5°F at resultant pressures."

NOTE: The margins of this amendment are marked with an asterisk or vertical lines to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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